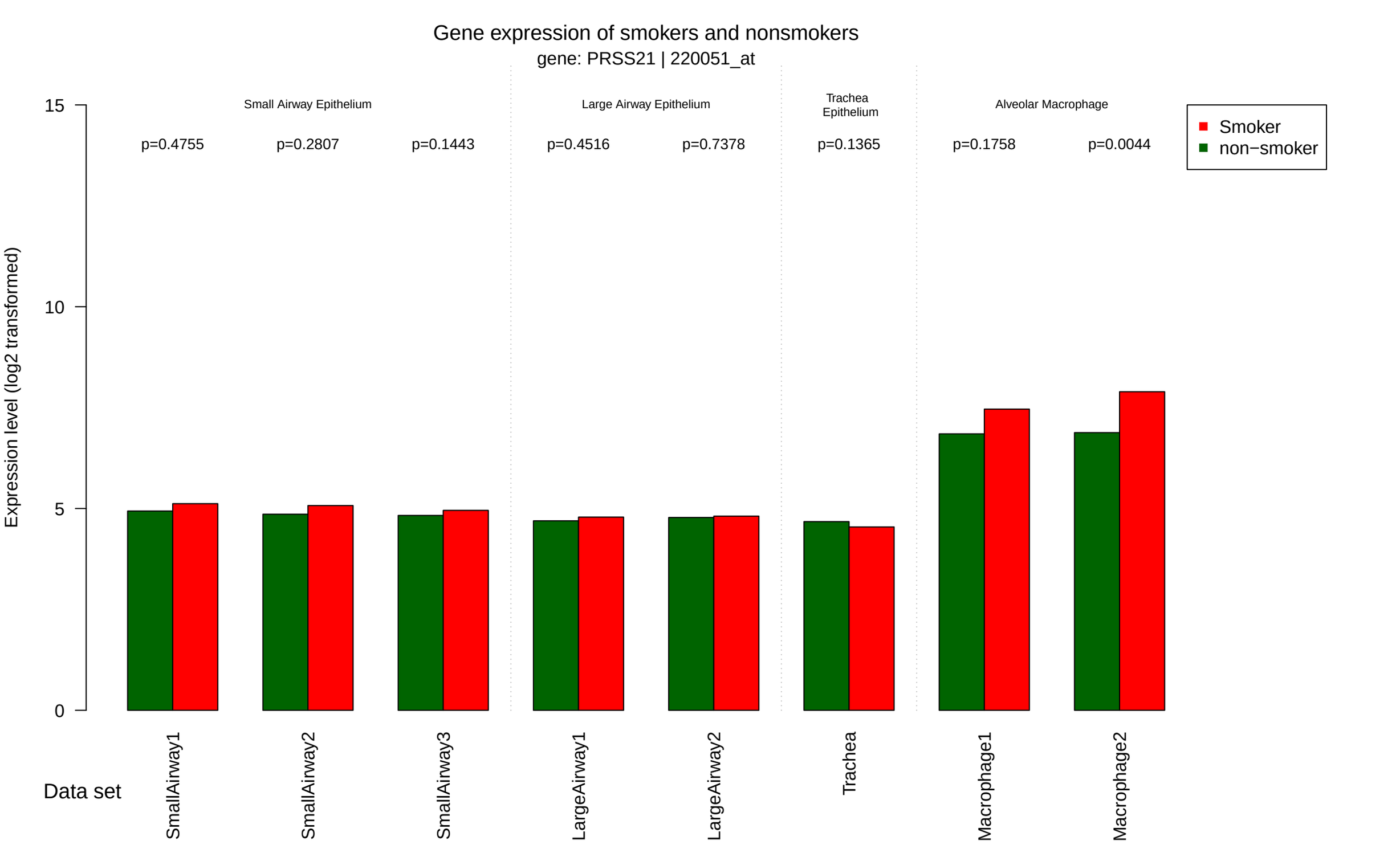
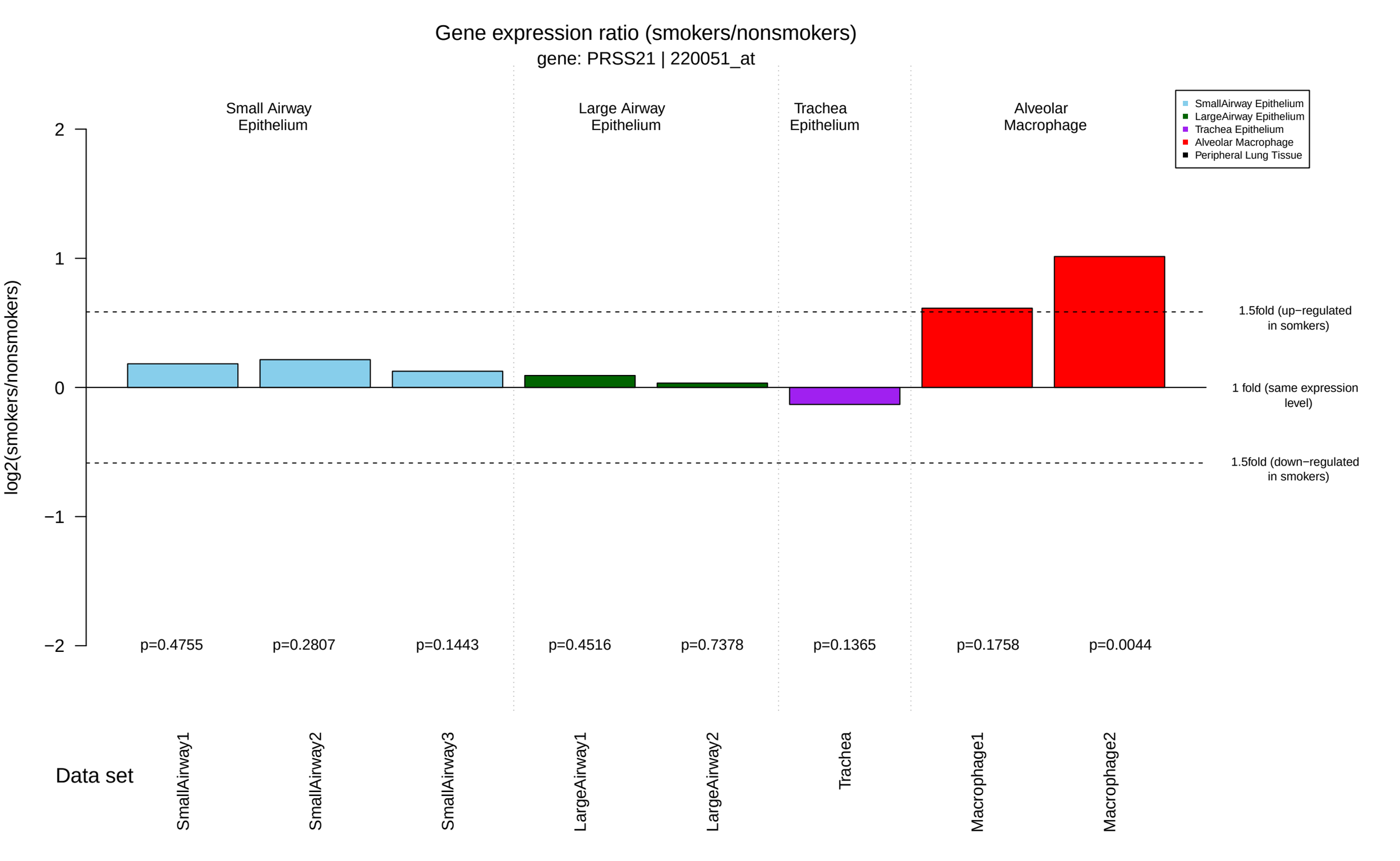
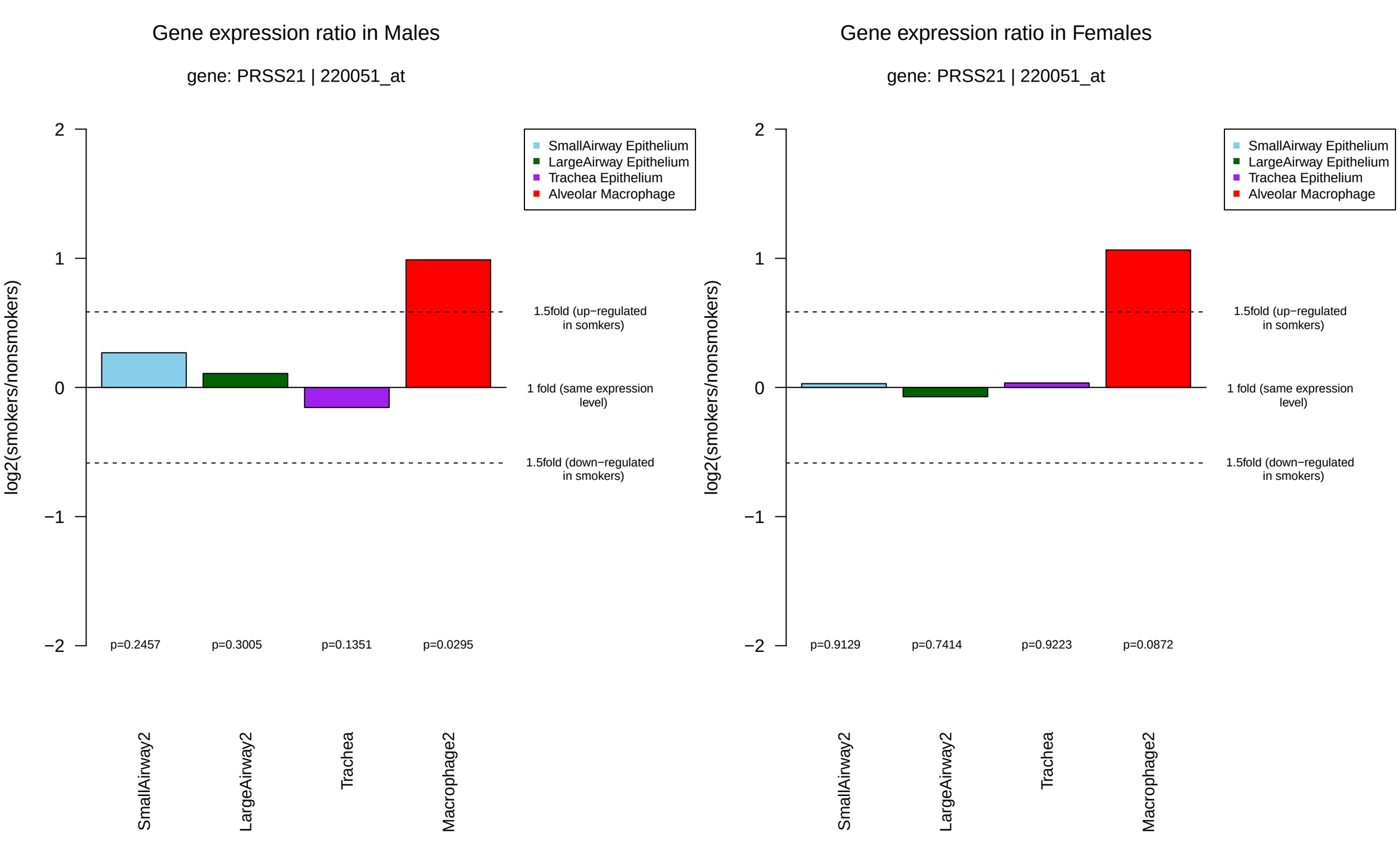
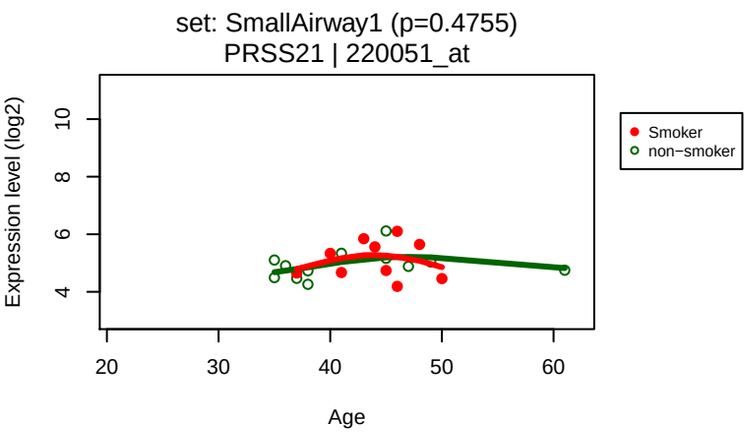
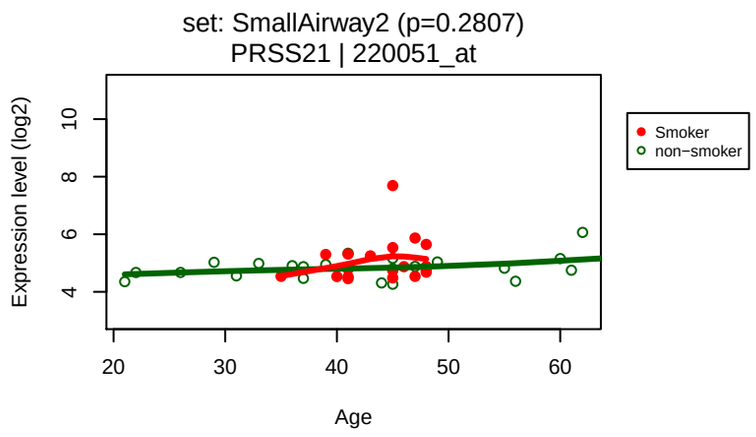
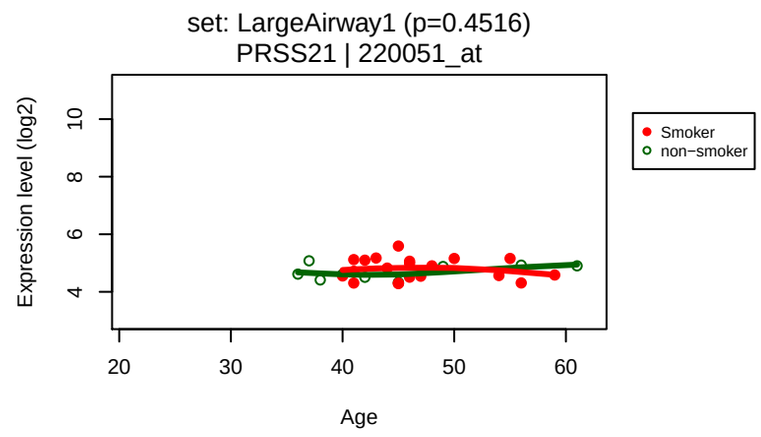

Gene expression of smokers and nonsmokers
gene: PRSS21 | 220051_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4755
p=0.2807
p=0.1443
p=0.4516
p=0.7378
p=0.1365
p=0.1758
p=0.0044
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

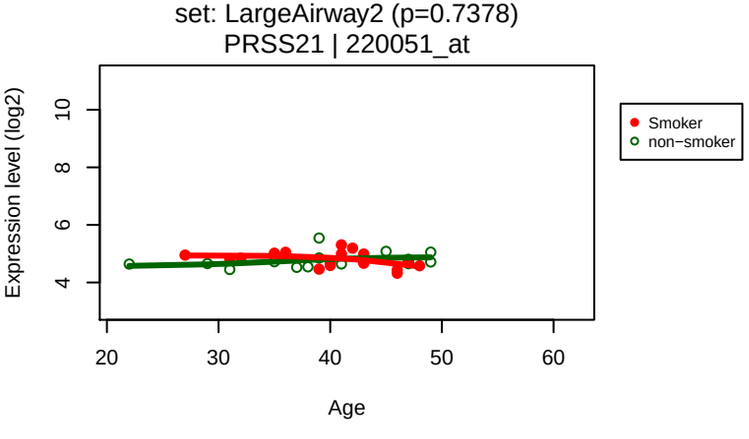

Gene expression ratio (smokers/nonsmokers)
gene: PRSS21 | 220051_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4755
p=0.2807
p=0.1443
p=0.4516
p=0.7378
p=0.1365
p=0.1758
p=0.0044
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

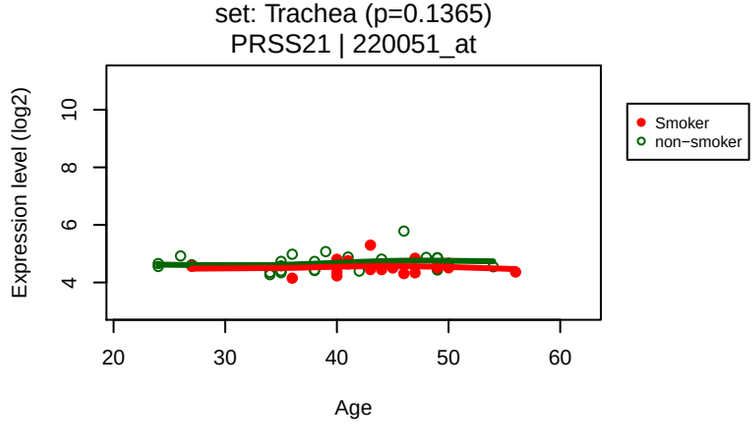

Gene expression ratio in Males
gene: PRSS21 | 220051_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2457
p=0.3005
p=0.1351
p=0.0295
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRSS21 | 220051_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9129
p=0.7414
p=0.9223
p=0.0872
SmallAirway2
LargeAirway2
Trachea
Macrophage2

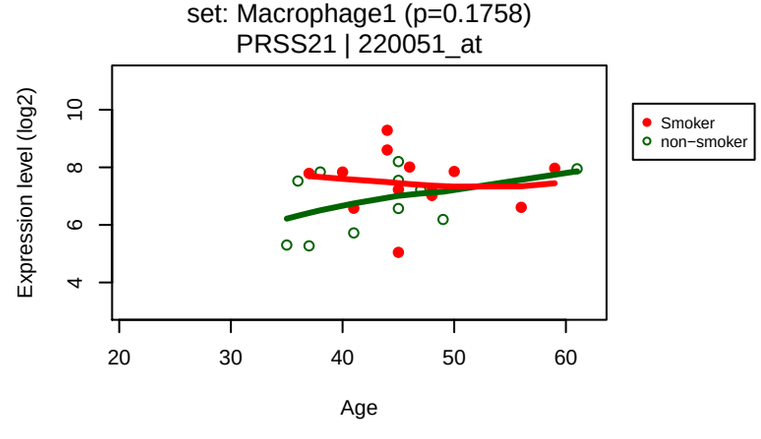

set: SmallAirway1 (p=0.4755)
PRSS21 | 220051_at
Expression level (log2)
Age
Smoker
non−smoker

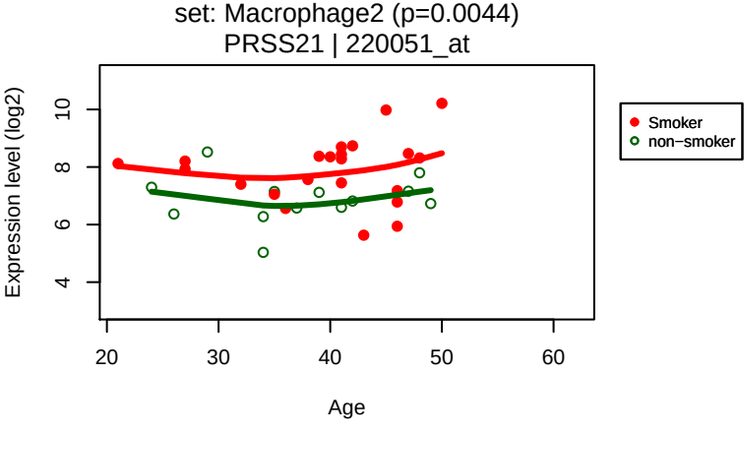

set: SmallAirway2 (p=0.2807)
PRSS21 | 220051_at
Expression level (log2)
Age
Smoker
non−smoker

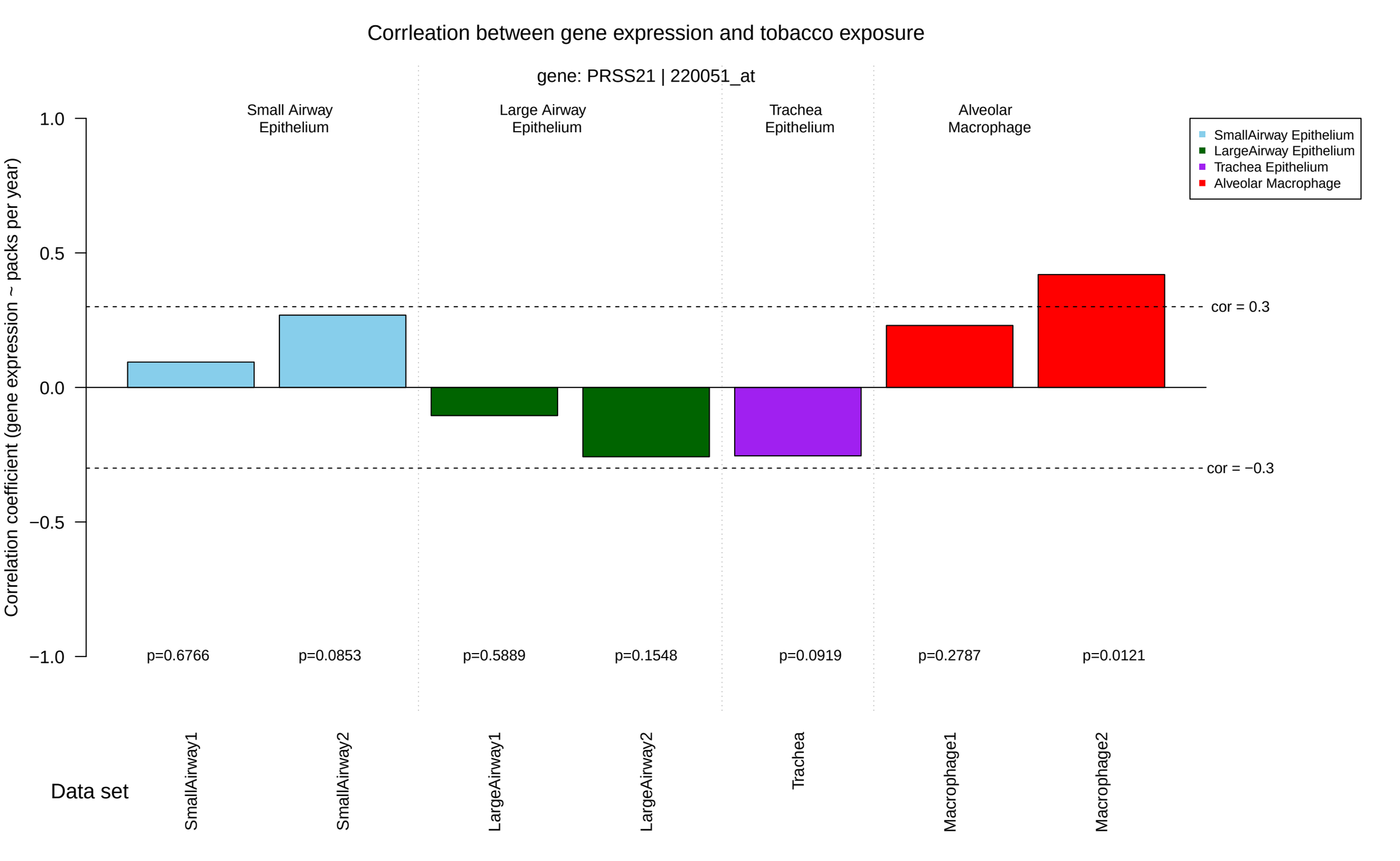

set: LargeAirway1 (p=0.4516)
PRSS21 | 220051_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7378)
PRSS21 | 220051_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1365)
PRSS21 | 220051_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1758)
PRSS21 | 220051_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0044)
PRSS21 | 220051_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRSS21 | 220051_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6766
p=0.0853
p=0.5889
p=0.1548
p=0.0919
p=0.2787
p=0.0121
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2